George Steven Muvva

georgestevenmuvva@gmail.com | +1 (785)-423-8498 |New York City, NY | LinkedIn

Professional Summary

AI / Machine Learning Engineer with 4 years of experience designing, developing, and deploying machine learning and deep learning solutions across healthcare and financial domains. Skilled in Python and modern ML frameworks including TensorFlow, PyTorch, and Scikit-learn, with hands-on experience in NLP and Transformer-based models (BERT). Experienced in building end-to-end ML workflows covering data preprocessing, feature engineering, model training, evaluation, deployment, and performance monitoring. Proficient in developing scalable ML systems using AWS, Spark, Docker, Kubernetes, and REST APIs, with a strong focus on production-ready, reliable, and maintainable AI solutions.

Technical Skills

- **Programming Languages:** Python, SQL, Java, C, Bash
- **Machine Learning & AI:** Scikit-learn, PyTorch, XGBoost, NLP, Transformers (BERT), SpaCy, NLTK
- **Deep Learning:** CNN, RNN, LSTM, GRU, Attention Mechanisms, Transfer Learning
- **Data Engineering & Big Data:** Pandas, NumPy, Apache Spark (PySpark), Kafka
- **Databases:** PostgreSQL, MySQL, MongoDB
- **Cloud & MLOps:** AWS (EC2, S3, Lambda, SageMaker), Docker, MLflow, CI/CD
- **Visualization & Tools:** Power BI, Tableau, Matplotlib, Seaborn, Git, Jupyter Notebook, VS Code
- **Concepts:** Feature Engineering, Model Evaluation, Hyperparameter Tuning, Data Pipelines, API Integration

Experience

Cardinal Health, New York City, NY| *AI Engineer|Healthcare Predictive Analytics Platform* **July 2025 - Present**

- Developed machine learning models for patient risk scoring and claims analytics using Python and Scikit-learn, improving prediction reliability through feature engineering and model tuning.
- Built NLP pipelines leveraging Transformers (BERT) and SpaCy to extract structured insights from unstructured clinical text data.
- Designed and deployed REST-based inference services using FastAPI to integrate ML predictions with downstream healthcare systems.
- Implemented scalable data processing workflows using Pandas, NumPy, and PySpark to support model training and evaluation.
- Containerized and deployed ML services using Docker and AWS, enabling reproducible training and consistent model versioning with MLflow.

Wells Fargo, Atlanta, GA| *Machine Learning Engineer | Banking Fraud Detection & Credit Risk Model* **May 2024 - June 2025**

- Built classification and anomaly detection models using XGBoost and Scikit-learn to support fraud detection and risk analysis use cases.
- Performed feature selection and model optimization to improve detection performance and reduce false positives.
- Developed real-time scoring APIs using Python and FastAPI for low-latency model inference.
- Designed ETL and data transformation workflows using PySpark and SQL for model training datasets.
- Applied model evaluation and explainability techniques, including ROC analysis and SHAP, to support validation and compliance needs.

Cognizant(CTS), Hyderabad, India| *Associate AI Engineer | Enterprise Data Intelligence Platform* **June 2021 - July 2023**

- Developed supervised learning models using Scikit-learn for forecasting and regression-based business problems.
- Performed data preprocessing and feature preparation using Pandas and NumPy across structured datasets.
- Built SQL-based data queries in MySQL to support analytics and model training workflows.
- Assisted in developing and deploying ML-backed APIs using Flask for internal applications.
- Created data visualizations and dashboards using Tableau to communicate analytical insights.

Projects

Intelligent Log Anomaly Detection System

- Designed an anomaly detection system to identify unusual patterns in application logs using Python and Scikit-learn.
- Engineered features from semi-structured log data (timestamps, error codes, response patterns) to improve detection accuracy.
- Implemented unsupervised models (Isolation Forest / One-Class SVM) to detect operational failures without labeled data.
- Built a FastAPI-based inference service to enable real-time log analysis and anomaly scoring.
- Visualized anomaly trends and system behavior using Matplotlib to support debugging and root-cause analysis.

Automated Resume Classification & Skill Extraction Engine

- Developed an NLP pipeline to classify resumes and extract key skills using Transformers (BERT) and SpaCy.
- Implemented text preprocessing, tokenization, and embedding workflows for handling unstructured resume data.
- Built multi-class classification models to categorize profiles based on role types and skill clusters.
- Designed rule-enhanced entity extraction for identifying technologies, experience levels, and domain expertise.
- Exposed the model via REST APIs using FastAPI to simulate integration with recruiting or ATS-style systems.

Education

University of Kansas | *Master of Science, Computer Science* **Aug 2023 - Dec 2025**